

Triage report of the 50 oldest open issues

openturns/openturns — branch `bug_hunt`

Automated triage by ox-alpha

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1 Introduction and method

This report classifies the 50 oldest *open* issues of the [openturns GitHub repository](#), ranging from issue #998 (January 2019) to issue #1585 (September 2020). Each issue body was read from the GitHub API; for ambiguous cases the full comment threads were also reviewed and cross-checked against the current source tree.

Each issue is assigned:

- a **category**: *actual bug*, *feature request*, or *other* (documentation work, usage questions, duplicates);
- an **effort estimate**: *easy* (contained change, at most a few files plus a test), *medium* (single subsystem, needs design or broad but mechanical edits), *hard* (new algorithms, API-wide refactors, platform-specific debugging), or *no more valid* (stale, duplicate, support-only, or already superseded in master).

Category assignment follows the *content* of the report rather than its GitHub labels, which proved unreliable: several items labelled **enhancement** describe genuine misbehaviours, while some items labelled **doc** hide behavioural questions.

2 Overview

Table 1: Effort estimate per category.

Category	Easy	Medium	Hard	No longer valid	Total
Bug	0	2	4	2	8
Feature	6	9	10	0	25
Other	8	5	1	3	17
Total	14	16	15	5	50

Key observations:

- Only 16% of the oldest backlog are confirmed bugs; the backlog is dominated by feature requests (50%) and documentation/questions (34%).
- No true bug of this vintage is an “easy” fix: the surviving bugs all involve numerical internals, drawing semantics or platform-specific behaviour.
- 5 issues (10%) are no longer actionable: one duplicate (#1396), two stale reports predating major renamings (#1231), superseded implementations (#1585) or answerable support questions (#1468, #1542).
- 14 issues (28%) qualify as easy wins, mostly documentation and small, well-contained API additions.

3 Detailed classification

#	Year	Title	Category	Effort
998	2019	Theory doc for interpolation methods missing	Other	Medium
1056	2019	SRC indices lack confidence bounds	Feature	Hard
1077	2019	Docstrings suggest kwargs are available	Other	Easy
1225	2019	Analytical gradients for kriging parameters estimation	Feature	Hard
1227	2019	Kernel smoothing: user-specified boundary conditions	Feature	Hard
1230	2019	Cannot override setDimension/setDescription from PythonDistribution	Feature	Medium
1231	2019	Cannot save NumericalPointWithDescriptionCollection	Bug	No longer valid
1237	2019	Anderson-Darling test for non-gaussian distributions	Feature	Hard
1239	2019	KernelSmoothing object from Parameters collection	Feature	Medium
1242	2019	Circular dependency ContinuousDistribution/DistributionImplementation	Bug	Hard
1243	2019	UserDefinedStationaryCovarianceModel ctor consistency	Feature	Easy
1244	2019	Performance of OrthonormalizationAlgorithm::getNodesAndWeights	Bug	Hard
1245	2019	Wrong comparison operator for interface classes	Bug	Medium
1246	2019	Distribution comparison improvement (semantic equality)	Feature	Hard
1250	2019	Iso-probabilistic transformation parameters	Bug	Hard
1256	2019	Move simulation draw methods to result classes	Feature	Hard
1262	2019	Storage backend for HistoryStrategy	Feature	Hard
1284	2019	ParetoFactory ML estimator not exact	Feature	Medium
1318	2019	Example of function with discrete input	Other	Easy
1320	2019	PythonCovarianceModel class	Feature	Hard
1322	2019	Fitting methods documentation	Other	Easy
1325	2019	Extract discrete/continuous parts of a distribution	Feature	Hard
1328	2019	GeneralLinearModelAlgorithm log-likelihood code question	Other	Medium
1353	2020	KrigingResult conditional variance as Function	Feature	Medium
1355	2020	Multiprocessing runs on Windows/Anaconda	Bug	Hard
1357	2020	DrawQQplot draws indistinguishable points	Feature	Easy
1358	2020	CorrelationAnalysis class undocumented	Other	Easy
1363	2020	DrawCorrelationCoefficients cannot draw SRC indices	Feature	Easy
1396	2020	MC generation with ComposedDistribution is column-wise	Other	No longer valid
1409	2020	SphericalModel must remove the a parameter	Feature	Medium
1410	2020	SphericalModel positive definiteness reference	Other	Easy
1424	2020	ProcessSample has no split method	Feature	Easy
1427	2020	Cannot extract part of a Sample by index list	Feature	Medium
1445	2020	Symbolic distribution arithmetics	Feature	Hard
1463	2020	Methods of Field have wrong name	Feature	Medium
1468	2020	Karhunen-Loeve expansion of a cyclic field	Other	No longer valid
1475	2020	Field has no arithmetic	Feature	Medium
1477	2020	KissFFT classes have no example	Other	Easy
1479	2020	Missing type check in CompositeDistribution ctors	Feature	Easy
1480	2020	Kernel smoothing improvements (refs, Silverman naming)	Other	Medium
1481	2020	Inadequate solver options are not signaled	Other	Medium
1488	2020	Distribution parameters in examples must be checked	Other	Hard
1489	2020	Mixture with Dirac does not draw its PDF correctly	Bug	Medium
1496	2020	No example of optimization config in EfficientGlobalOptimization	Other	Easy

#	Year	Title	Category	Effort
1511	2020	LeastSquaresMethod doc issues	Other	Medium
1542	2020	Conditional GP simulation and matrix compression	Other	No longer valid
1554	2020	Weighted samples in KernelSmoothing	Feature	Medium
1582	2020	Inconsistent stochastic process definitions in theory	Other	Easy
1583	2020	Uneasy to configure the bounds of a Graph	Feature	Easy
1585	2020	KS fitting test p-value takes random values	Bug	No longer valid

4 Category analysis

4.1 Actual bugs (8 issues)

- #1231 – Cannot save NumericalPointWithDescriptionCollection (no longer valid).** Uses pre-1.14 API names that no longer exist; not reproducible against any supported version. Should simply be closed after a quick sanity check.
- #1242 – Circular dependency of ContinuousDistribution (hard).** Design flaw acknowledged by maintainers: remove the `ContinuousDistribution/DiscreteDistribution` layers. Touches the whole distribution hierarchy.
- #1244 – Performance of OrthonormalizationAlgorithm (hard).** Numerical robustness/performance of recurrence-coefficient computation for extreme parameters; requires algorithmic expertise.
- #1245 – Wrong comparison operator for interface classes (medium).** `ot.SobolSequence(2) == ot.HaltonSequence(2)` returns true because interface `operator==` compares only the top-level implementation type. Fix requires virtual dispatch in several hierarchies.
- #1250 – Iso-probabilistic transformation parameters (hard).** Parameter reporting of marginal/copula transformations is inconsistent; needed for FORM sensitivities, so changes must be made carefully.
- #1355 – Multiprocessing on Windows/Anaconda (hard).** Output printed once per worker with `n_cpu > 1`; Windows-specific process/spawn semantics of `PythonFunction`; hard to reproduce and debug outside Windows.
- #1489 – Mixture with Dirac does not draw its PDF (medium).** The discussion concludes that mixed distributions should render their atoms as arrows whose height relates to the dominant density mode; requires drawing-machinery work.
- #1585 – KS fitting test p-value random (no longer valid).** Declared invalid (Monte-Carlo p-value by design). The follow-up action item (raise `FittingTest-KolmogorovSamplingSize`) has meanwhile been superseded in master by an adaptive Lilliefors scheme (`FittingTest-LillieforsMinimumMaximumSamplingSize, ...Precision`). Close as resolved/superseded.

4.2 Feature requests (25 issues)

Hard items are substantial algorithmic or API-wide projects: SRC confidence bounds (#1056), analytical kriging gradients (#1225), kernel-smoothing boundary specification (#1227), Anderson-Darling for non-gaussian laws (#1237), semantic distribution equality (#1246), moving draw methods to result objects (#1256), a storage backend for `HistoryStrategy` (#1262), `PythonCovarianceModel` (#1320), discrete/continuous decomposition (#1325), symbolic distribution arithmetics (#1445).

Easy, well-contained additions:

- #1583 convenience axis-bound setters for `Graph` (`setXMin/setXMax/setYMin/setYMax`);

- #1424 `ProcessSample::split`, mirroring `Sample::split`;
- #1363 proper y-axis labelling when plotting SRC indices;
- #1357 QQ-plot point style so outliers remain visible;
- #1243 constructor consistency for `UserDefinedStationaryCovarianceModel`;
- #1479 flattening/type checks in `CompositeDistribution` constructors.

4.3 Other: documentation, questions, duplicates (17 issues)

Eight easy documentation tasks (#1077 kwargs ambiguity in ctor docstrings, #1318 discrete-input function example, #1322 fitting-methods notes, #1358 `CorrelationAnalysis` reference pages, #1410 `SphericalModel` positive-definiteness note, #1477 `KissFFT` example, #1496 EGO configuration example, #1582 theory-page notation consistency), five medium ones (#998, #1328, #1480, #1481, #1511) and one hard sweep over examples (#1488). Three require no code change: #1396 (duplicate), #1468 and #1542 (support questions answered in-thread).

5 Action plan for the easy fixes

The following easy code fixes were selected to start with, ordered by value:

1. **#1583** `Graph`: add `setXMin/setXMax/setYMin/setYMax` implemented on top of `setBoundingBox`, switching off automatic bounding box; document them; add a test.
2. **#1424** `ProcessSample`: add `split(index)` mirroring `Sample::split`; add a test.
3. **#1363** `SobolIndicesAlgorithm`: allow a dedicated y-axis label so SRC indices are no longer plotted as “correlation coefficient”.
4. **#1410** `SphericalModel` help page: document that positive definiteness is guaranteed only for spatial dimension ≤ 3 , with the Stein (1999) reference.

Remaining easy items (mostly examples/docstrings) can then be processed in batches; the two invalid bug reports should be closed upstream.

6 Recommendations

- Close #1231, #1396, #1468, #1542 and re-triage #1585 (superseded by the Lilliefors implementation).
- Fix label hygiene: #1489 and #1585 are bugs filed under other labels; #1231 filed as enhancement.
- Batch the eight easy documentation issues into a single “documentation day” PR series.
- The medium bugs #1245 and #1489 are good next candidates after the easy batch.

7 Proposed qualifiers for the remaining open issues

Five of the triaged issues should be closed outright (#1231 stale, #1396 duplicate, #1468 and #1542 answered support questions, #1585 superseded by the Lilliefors implementation); they receive no replacement qualifiers. The 45 remaining issues are re-labelled below using a coherent three-axis scheme: **type** (bug, enhancement, doc, question), **area** and **effort** (effort:easy, effort:medium, effort:hard). Issues #1583, #1424, #1363 and #1410 additionally receive **fix-ready** once the corresponding patches are submitted upstream.

#	Proposed qualifiers
998	doc, theory, effort:medium
1056	enhancement, sensitivity, effort:hard
1077	doc, API, effort:easy
1225	enhancement, surrogate, effort:hard

#	Proposed qualifiers
1227	enhancement, probas, effort:hard
1230	enhancement, API, python, effort:medium
1237	enhancement, probas, effort:hard
1239	enhancement, API, probas, effort:medium
1242	bug, probas, refactor, effort:hard
1243	enhancement, API, process, effort:easy
1244	bug, performance, probas, effort:hard
1245	bug, API, effort:medium
1246	enhancement, probas, API, effort:hard
1250	bug, reliability, API, effort:hard
1256	enhancement, API, refactor, effort:hard
1262	enhancement, performance, effort:hard
1284	bug, probas, effort:medium
1318	doc, example, effort:easy
1320	enhancement, surrogate, python, effort:hard
1322	doc, probas, effort:easy
1325	enhancement, probas, API, effort:hard
1328	question, surrogate, needs review, effort:medium
1353	enhancement, surrogate, effort:medium
1355	bug, performance, python, windows, effort:hard
1357	enhancement, graph, effort:easy
1358	doc, sensitivity, effort:easy
1363	enhancement, sensitivity, graph, effort:easy, fix-ready
1409	enhancement, API, surrogate, effort:medium
1410	doc, surrogate, effort:easy, fix-ready
1424	enhancement, process, API, effort:easy, fix-ready
1427	enhancement, sample, API, effort:medium
1445	enhancement, probas, effort:hard
1463	enhancement, API, process, refactor, effort:medium
1475	enhancement, process, effort:medium
1477	doc, example, signal, effort:easy
1479	enhancement, probas, effort:easy
1480	doc, probas, API, effort:medium
1481	doc, optimization, effort:medium
1488	doc, example, effort:hard
1489	bug, graph, probas, effort:medium
1496	doc, example, optimization, effort:easy
1511	doc, API, effort:medium
1554	enhancement, probas, effort:medium
1582	doc, theory, process, effort:easy
1583	enhancement, graph, API, effort:easy, fix-ready

New qualifier vocabulary introduced here: **theory**, **example**, **refactor**, **performance**, **process**, **sample**, **signal**, **python**, **windows**, **effort:*** and **fix-ready**; all other qualifiers reuse existing repository labels. The old **difficulty:easy**/**difficulty:medium** labels map to **effort:easy**/**effort:medium**.

8 Second batch: next 50 open issues

A second pass classified the following 50 oldest open issues (#1587–#2164, September 2020 – October 2022) with the same method. Issues fixed during the working session are marked “(fixed)”.

Table 4: Effort estimate per category, second batch.

Category	Easy	Medium	Hard	Total
Bug	1	1	4	6
Feature	3	15	4	22
Other	10	9	3	22
Total	14	25	11	50

#	Title	Category	Effort
1587	GridLayout subplots	Feature	Medium
1595	Undocumented ResourceMap keys	Other	Hard
1596	Mixture PDF hides discrete part	Bug	Easy (fixed)
1601	Sobol'/Halton sequences start at zero	Feature	Hard
1678	Process event example vs theory	Other	Easy (fixed)
1700	Kriging setNoise vs setNuggetFactor	Other	Medium
1703	P1 gram / KL decomposition performance	Bug	Hard
1707	ANCOVA indices outside [0,1]	Other	Easy (fixed)
1720	Distribution square root hangs	Bug	Hard
1722	StandardEvent full-length example	Other	Medium
1730	Sample.erase() with a set of indices	Feature	Easy (fixed)
1734	Minimum volume level sets for several levels	Feature	Medium (fixed)
1741	Factories do not describe their method	Other	Medium
1757	Finite difference classes without example	Other	Medium
1762	FunctionalChaosSobolIndices ResourceMap key	Other	Easy
1765	PCG instead of Mersenne Twister	Feature	Hard
1769	GridLayout limited features	Feature	Medium
1779	Draw methods ignore default palette	Feature	Medium
1780	OpenTURNPythonFunction example	Other	Medium
1786	EGO expected improvement accessor	Feature	Medium
1841	AbdoRackwitz ignores error thresholds	Bug	Medium (fixed)
1850	Missing distribution factories	Feature	Hard
1852	Chaos moments validity check	Feature	Medium
1853	Merging distributions through copulas	Other	Easy
1873	GP sampling instability	Bug	Hard
1875	PythonDistribution automatic CDF	Feature	Medium
1886	Jumbled high-dimensional PlotDesign	Feature	Medium
1888	numpy.where on Point/Sample	Feature	Hard
1892	Sobol' with low discrepancy example	Other	Easy
1893	FORM versus discrete variables	Feature	Medium
1899	ResourceMap key reference consistency	Other	Medium
1924	SpaceFillingMinDist wrong doc or code	Other	Medium
1928	Default finite difference formula doc	Other	Easy
1937	FunctionalChaosSobolIndices.draw()	Feature	Easy (fixed)
1948	ProductFunction constructor too rigid	Feature	Medium
1951	discretizeRow undocumented	Other	Easy
1957	OptimalLHS const correctness	Feature	Medium
1960	SimulationSensitivityAnalysis examples	Other	Medium
1992	HMatrix theoretical documentation	Other	Hard
1996	FunctionCollection slicing	Feature	Medium (fixed)
1999	Collection classes doc issue	Other	Easy
2000	Collection setter doc consistency	Other	Easy
2078	Student estimation inaccurate df ₂	Bug	Hard
2080	Wrong cross reference in KL SVD doc	Other	Easy
2114	Theory/API cross links generalization	Other	Hard
2118	Cloud point size control	Feature	Medium
2119	ComposedFunction left/right accessors	Feature	Easy (fixed)
2127	Navigation bar improvements	Other	Medium
2128	enumerate API renames	Feature	Medium
2164	BoxCoxFactory BasisCollection deprecation	Feature	Medium

Notable differences with the first batch: no duplicate or stale report, and a larger share of feature requests. The GP/Kriging related items (#1700, #1703, #1786, #1873) were deliberately parked because these classes evolve in parallel pull requests, as well as the API renames and deprecations (#1601, #2128, #2164) which require an upstream policy decision.

9 Session outcome

Sixteen issues of the two batches received a fix during the session, one commit each:

- #1363 SRC y-axis label, #1410 SphericalModel note, #1424 ProcessSample.split, #1583

Graph bounds setters;

- #1318 discrete input example, #1322 fitting estimation notes, #1477 KissFFT example, #1496 EGO solver configuration, #1582 process mean notations;
- #1357 visible plot points, #1479 nested CompositeDistribution;
- #1596 arrows for any discrete component, #1678 process Monte Carlo example, #1707 AN-COVA indices range, #1730 erase a set of indices, #1937 chaos Sobol indices drawing, #2119 ComposedFunction accessors;
- #1243 single matrix constructor, #1554 weighted kernel smoothing, #1734 multi-level minimum volume level sets, #1841 AbdoRackwitz convergence reason, #1996 collection slicing.

Three further issues are verified fixed upstream or superseded (#1077, #1358, #1427, plus #1245 in another pull request) and five of the first batch should be closed outright (#1231, #1396, #1468, #1542, #1585).

10 Third batch: next 50 open issues

Third pass over the following 50 oldest open issues (#2178–#2537, December 2022 – May 2024). Issues fixed during the session are marked “(fixed)”, issues already resolved upstream are marked “superseded”.

Table 6: Effort estimate per category, third batch.

Category	Easy	Medium	Hard	Superseded	Total
Bug	3	3	2	2	10
Feature	4	8	5	0	17
Other	13	9	1	0	23
Total	20	20	8	2	50

#	Title	Category	Effort
2178	HaarWavelet/Fourier signal examples	Other	Medium
2201	FiniteBasis not exposed to Python	Feature	Easy
2228	Deprecate Factory.buildAs* methods	Feature	Medium
2247	getResult returns wrong result class in docs	Other	Easy
2248	NormalCopula.getCorrelation documentation	Other	Easy
2249	Switch COBYLA to PRIMA implementation	Feature	Hard
2253	getBasisSizeFromTotalDegree example bug	Other	Easy
2260	Sobol indices documentation versus code	Other	Medium
2280	drawObservationsVsInputs misleading plot	Other	Medium
2294	Polynomial chaos documentation rewrite	Other	Hard
2319	GP Gibbs sampling with output dimension ≥ 1	Bug	Hard
2331	setConfidenceLevel undocumented ResourceMap key	Other	Easy
2333	Aggregated Sobol indices from a PCE	Feature	Medium
2334	Exact Ishigami coefficients in C++ tests	Feature	Easy
2335	LeastSquaresStrategy API help issues	Other	Easy
2355	Full/active parameters documentation	Other	Medium
2365	Kriging trend coefficients ordering doc	Other	Medium
2370	Numerical methods section on the website	Other	Medium
2376	Principal component analysis class	Feature	Hard
2378	ExpectationSimulationAlgorithm with an experiment	Feature	Medium
2379	SobolSimulationAlgorithm bootstrap keys doc	Other	Easy
2380	Sobol indices distribution from PCE by bootstrap	Feature	Medium
2383	Model selection example for step B	Other	Medium
2385	LeastSquaresMethod doc wrongly tied to PCE	Other	Medium
2390	Relative default quantile epsilon	Feature	Medium
2400	PenalizedLeastSquares ridge completion	Other	Medium
2401	QuantileMatchingFactory robustness	Bug	Medium

#	Title	Category	Effort
2407	PythonFunction setNCpus after construction	Feature	Medium
2408	LeastSquaresExpansion/IntegrationExpansion doc	Other	Easy
2409	SparseMethod purpose and example	Other	Easy
2428	FieldImplementation.drawMarginal missing colorbar	Bug	Medium
2432	Point sum dimension mismatch message	Bug	Easy (fixed)
2436	Error messages should state their origin	Feature	Hard
2440	Copula FittingTest normalization clarification	Other	Easy
2444	Multivariate goodness of fit test for copulas	Feature	Hard
2445	Suboptimal TruncatedDistribution.drawPDF range	Bug	Medium
2465	Unused dNonCentralStudentAlt0 function	Other	Easy
2482	Legend corner API redesign	Feature	Medium
2483	Combining OpenTURNS graphics with Matplotlib	Other	Easy
2484	CorrelationAnalysis API renaming	Feature	Medium
2485	Matrix to Sample conversion	Feature	Easy
2494	getCoefficientsStandardErrors formula	Other	Easy
2498	QQ plot ignores largest observation	Bug	Easy (fixed)
2502	Covariance out of bounds message	Bug	Easy (fixed)
2509	Gumbel default location parameter	Feature	Easy
2512	Sphinx substitution for experimental warning	Other	Easy
2521	Normal.computeProbability infinite bounds	Bug	Superseded
2522	MetaModelAlgorithm failure with -O3 on POWER	Bug	Hard
2533	Specialized simulation getResult classes	Feature	Hard
2537	ChiSquared factory discrete mismatch	Bug	Superseded

The GP related item #2319 remains parked like the rest of that area. Two issues were found already fixed in master (#2521 infinite bounds in Normal::computeProbability, #2537 discrete ChiSquared through a factory) and should be closed. The session fixed three contained bugs: the swallowed dimension mismatch message of Point additions, the misleading out of bounds message of symmetric matrices, and the dropped largest observation of QQ plots.